

Animal semantics

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Abstract

Semantics, the study of *meaning*, plays a central role in research on animal communication, with the goal of uncovering and understanding homologies or analogies to the way that meaning is communicated in human language. One challenge is that, unlike human language (which can be studied with direct introspective judgments), animal meaning can only be investigated with indirect measures. Recent work has shown that concepts and tools from formal semantics—built for human language—can nevertheless be successfully applied to the study of animal meaning. In this chapter, we use a formal semantic approach to define meaning, semantic denotation and pragmatic inferences in animal communication, lay out their relevance to several key methodologies, and describe principles for investigating the semantics-pragmatics and syntax-semantics interfaces.

Key points

- The meaning of an animal signal can be defined extensionally in terms of features of the contexts in which it is used.
- The semantic denotation is the component of meaning that remains constant across its contexts of use; pragmatics pertains to the components of meaning that depend on context.
- Receiver reaction is a consistent component of meaning, but identifying the denotation of a signal (as opposed to pragmatic inferences) requires analyzing the signal and reaction in context.
- Game-theoretic principles (including informativity and relevance) can be used to disentangle denotation and pragmatic inference in animal semantics.

1. Introduction

Meaning has played a central role in research on animal communication from the foundational work on the topic (e.g. Seyfarth et al. 1980), to recent collaborations between biologists and linguists (e.g. Schlenker et al. 2016, Berthet et al. 2023). In much of this work, one implicit long-term goal is to uncover and understand homologies or analogies to the way that meaning is communicated in human language (Hauser et al. 2002). This endeavor faces the challenge, however, that the relatively direct manner in which human language can be studied (direct introspective judgments, Marantz 2005) is impossible to conduct on non-human subjects, for which only indirect measures can be collected.

Recent work, however, has shown that the framework and analytical tools of formal semantics—as developed for human language—can nevertheless be productively applied to animal communication, providing a principled basis on which meaning for humans and non-human animals can be compared. In the present article, we present several key elements of the enterprise of formal semantics as applied to animal communication. Along the way, we observe how various methodological approaches relate to the approach and highlight some new generalizations and questions that it has motivated.

2. What is meaning?

What counts as meaningful behavior? Here we adopt a relatively broad characterization of meaningful behavior and the meaning it carries. Meaningful behavior is thus not restricted to only symbolic parts of a code (cf. Grice 1957), but encompasses signals more generally—sources of information plastically produced at a cost with a communicative function (adapted from Bradbury & Vehrencamp, 2011). Applied to human language, this characterization of meaningful behavior encompasses a larger set of behaviors than what is generally considered language proper, including, for example, leaving a pile of dirty clothes in the middle of the room as a signal to another person in the household (or to oneself) to do the laundry. Likewise, we characterize meaning broadly as the set of inferences that can be drawn from a signal in a particular context. The sentence ‘It’s raining’ thus expresses a core meaning involving falling water (the *semantic denotation*, below), but also, depending on context, may express the meaning that the laundry should be taken inside, or that one should take an umbrella (the *pragmatic inferences*, below).

Because the study of animal meaning relies on observations (as opposed to introspective judgments), any first-pass definition of meaning is necessarily *extensional*, that is, relating to features of the actual world at a given time (Fitting 2022). Features can relate to a variety of different circumstances of the context in which a signal is produced, including: external events, permanent traits of the signaller, transient states of the signaller, the receiver’s behavioural reaction, and the signaller’s behaviour. We correspondingly define the meaning of a signal as the set of features of circumstances that appear at a rate greater than chance across the signal’s occurrences (Berthet et al. 2023). (Observe that under this definition, a feature may co-occur frequently with a given signal, but this is irrelevant to the signal’s meaning if the feature also appears frequently with other signals.)

2.1 The semantic denotation

In the semantics of human language, the *denotation* of a word or utterance is defined as its stable semantic contribution: that component of meaning that remains constant across its many contexts of use (cf. Katz & Fodor 1963). Thus, while ‘It’s raining’ may sometimes express information about laundry and sometimes not, what is common across all uses of the utterance is the presence of falling water.

For animal communication, the semantic denotation can thus be defined as the largest set of meaningful features of circumstances that appear across all occurrences of the signal (Berthet et al. 2023). For example, the alarm calls of vervet monkeys (*Chlorocebus pygerythrus*) are produced in response to specific classes of predators but also during aggression events; what appears to be common to both situations is a similar emotional state of the caller—a feature of the transient state of the signaller (Price et al., 2015; but see Schamberg et al. 2018). In contrast, it has been argued that the alarm ‘hoos’ of chimpanzees

(*Pan troglodytes*) denote ambush threats—a feature of the external events. Specifically, alarm ‘hoos’ are emitted in the presence of snakes, but emission can be suppressed when the receivers are already aware of the situation, suggesting that the calls do not denote traits or behavior of the caller or behavior of the recipient (Crockford et al. 2012, 2017; see also Girard-Buttoz et al. 2020). Signals can also denote receiver behaviors; for example, in a group of wild chimpanzees, all instances of the ‘present climb on’ gesture resulted in the receiver climbing on the signaller (Hobaiter & Byrne 2014).

2.2 Pragmatic inferences

Pragmatics, addressed further in Chapter XX of this volume, pertains to those components of meaning that depend on context. Saying ‘I need to pee!’ to the driver of a car is a request to find a rest stop, while saying the same sentence to a roommate taking a long shower is a request to free the bathroom. On the current perspective, both of these inferences are part of the meaning of the signal, but they only arise in certain contexts. Pragmatic inferences—also called *implicatures* in the linguistic literature—can be thought of as involving reasoning, deriving further information about a state of affairs based on the fact that a given meaningful signal was emitted in a particular context.

For animal communication, we define pragmatic inferences as the set of meaningful features of circumstances that always appear when a signal is uttered *in the presence of a given contextual feature*. Investigating animal pragmatics thus involves identifying the contextual features that may be involved in pragmatic reasoning in the communication of different species. These features may be low-level, perceptually-available properties of a context. For example, female putty-nosed monkeys (*Cercopithecus nictitans*) react to alarm calls differently depending on other environmental cues that may further identify the cause of alarm, such as the noise of a falling tree (Arnold & Zuberbühler 2013; see also Arnold & Bar-On 2020); similarly, when hearing alarm calls that do not denote the location of a threat, listeners can infer the location of the predator by looking at the gaze direction of the signaller (Davidson et al. 2014). Contextual features may also include properties that are not directly observable. These include representations of social structure; for example, chimpanzees react differently to aggressive barks depending on whether or not the signaller has a close social bond to a previous opponent of the receiver (Wittig et al. 2014). Non-directly observable contextual features may also include higher level representations of the beliefs or intentions of others, such as others’ knowledge; in particular, wild chimpanzees have been shown to adapt their communicative behavior depending on the apparent epistemic state of conspecifics (Crockford et al. 2012).

3. Meaning and receiver behavior

A fundamental function of communication is to effectuate change in another individual. In some cases, this is direct behavioral change (‘I need to pee!’ to get one’s roommate out of the bathroom); it can also be a less visible change, such as a change of informational (or emotional) state of the receiver (‘Did you know that ...’), although even in these cases, there may be a communicative function to produce a particular behavioral response, possibly delayed in time (‘You look nice!’ to increase positive interactions from the receiver in the future). Importantly, this does not mean that receiver behaviour is necessarily part of the semantic denotation of a communicative signal, but it does mean that receiver

behavior will be one of the most consistent inferences (whether semantic or pragmatic) carried by a message.

Several ethological methodologies are founded on this insight. Most overtly, Hobaiter & Byrne (2014) define an *Apparently Satisfactory Outcome* (ASO) as “an outcome that result[s] in the cessation of communication and that represent[s] a plausible desire on the part of the signaler,” and take ASOs “to indicate the signalers’ intended meanings.” The ASO methodology can be carried out on any signal modality (acoustic, gestural, etc.). Additionally, *playback experiments*, in which an audio recording of a call is played to other animals, also use observed receiver behavior as the measured variable, and thus the only recorded meaning of the signal (Fischer et al., 2013). Unlike the ASO, playback experiments are limited to audio signals for technical reasons. The role of context is different in the two methodologies: the ASO is based on observed behavior so the signals are clearly ethologically felicitous, and a large set of features of context can be observed, including signaller behavior; in contrast, playback experiments in principle allow more control over contextual factors including testing relatively rare signals or contexts, but risks resulting in ecologically inappropriate signals, yielding uninterpretable receiver behavior. Both methodologies are united in using receiver behavior as the only measured component of meaning, which, as we have seen, is not necessarily the core *denotation* of the signal (Hobaiter et al. 2022). This may lead to unclear conclusions for non-imperative signals (e.g. emotional or referential calls) that could result in a high diversity of receiver reactions. Nevertheless, pragmatic inferences do not exist in isolation; they are fundamentally linked to the semantic denotation, so analyzing the inferences of a message (including receiver behavior) *in context*, armed with a theory for the interaction of meaning and context, allows one to triangulate on what the denotation could be.

4. The semantics-pragmatics interface

Disentangling denotation and pragmatic inference has been a central part of recent work in animal semantics. While there is no magic algorithm to follow, there are well-developed theories about meaning and context that provide a framework in which to build hypotheses. These have in several cases revealed new insights or generalizations about animal communication.

The relation between semantics and pragmatics can be thought about in game-theoretic terms, in which a message with denotation m is employed to communicate information i , where m and i are not necessarily identical, due to limits on the set of possible messages (e.g. a finite inventory or a cost on message complexity). Identifying the intended i (for the receiver) or the optimal m (for the signaller) ultimately involves reasoning about what is said, what is not said, and when it is said, in the ecological context in which the act of communication takes place (Grice 1975). (We note that these principles do not necessarily rely on any theory of mind; see Berthet et al. 2023 for more discussion.)

A first clear implementation of these principles involves *informativity*, and compares a signal that was used to a more specific signal that could have been used instead. For example, if a person says to their spouse, ‘A child is playing in the driveway,’ there is an inference that the child in the driveway is not their own child Steven. This inference is not part of the denotation of the sentence—after all, the sentence is still true if Steven *is* playing in the driveway. Rather, the inference arises by competition with a more specific alternative: if the child were Steven, it would have been more efficient and more informative to simply say ‘Steven is playing in the driveway.’ The fact that the more informative message was not

chosen leads to the inference that it is not true. Following the same logic, Schlenker et al. (2016) argue that a similar situation exists for alarm calls of various species—that one alarm call in their repertoire is more informative (i.e. more specific) than another, giving rise to informativity-based inferences. For example, male Campbell's monkeys have a call 'hok' that is typically associated with aerial threats and a call 'krak' that is typically associated with terrestrial threats (Ouatara et al. 2009); Schlenker et al. (2014) argue that 'krak' is in fact a general alarm call, whose denotation encompasses both aerial and terrestrial threats. As a consequence, the association with terrestrial threats is a pragmatic inference arising from competition with the more informative 'hok', which specifically denotes aerial threats. In other words, 'hok' means 'there is an aerial threat' and 'krak', 'there is a general alert' which, through pragmatic inferences, is enriched to 'there is a general alert but not an aerial one'.

A second implementation of these principles involves *relevance*, especially with respect to the timing of the signal (Why was this mentioned and why now?). In particular, Schlenker et al. (2016) describe a principle of *urgency*, which says that urgent information (such as information related to threats) should be communicated as soon as possible. They use this principle to explain the meaning of the 'hack' calls of putty-nosed monkeys, which generate different inferences depending on where they appear in a call sequence: when used alone or at the beginning of a call sequence, they indicate (urgent) information about aerial predation; when following other calls in a call sequence, they convey information about non-ground movements. Similarly, in the long alarm sequences of titi monkeys, urgent information about the predatory event is conveyed at the beginning of the sequence, while less-urgent information about group identity is conveyed later (Narbona Sabaté et al. 2022).

Several further remarks are in order. First, deceptive communicative behavior has been claimed for a variety of species (e.g. Flower et al. 2014; Kean et al. 2017). On the one hand, such behavior will interfere with rote application of the definition of the semantic denotation provided above, as deceptive messages are uttered precisely when their meaning is *not* true. Nevertheless, deception actually enters perfectly into theories of the semantics-pragmatics interface (e.g. bluffing does not require a theory of mind; cf. Kuhn 1950); and measuring inferences (including notably receiver behavior) still allows one to home in on what the denotation is or could be. (Note, for example, that playback experiments are fundamentally based on deception, yet of course give useful information to the researcher.)

Finally, even in human language, the division between semantics and pragmatics is not always clear-cut; indeed, in some cases, what were originally pragmatic inferences may in time be conventionalized as part of the semantic denotation. For example, a common pattern in language change is semantic narrowing, in which the denotation of a word becomes more specific; Horn (1984) argues that such cases can often be seen as conventionalization of a pragmatic inference arising either from informativity ('cow' coming to denote specifically females, due to competition with 'bull') or relevance ('drink' coming to mean 'drink alcohol', as a relevant subcase; e.g. 'I don't drink'). A similar situation may well exist for animal communication. Notably, *ritualization*, in which an originally instrumental behavior changes function to become a communicative behavior (Halina et al. 2013), may be viewed as the shift from pragmatics to semantics in ontogeny or even phylogeny.

5. Compositional semantics (the syntax-semantic interface)

A combination of signals can be classified as either a compositional combination—the meaning is derived in a systematic way from the meanings of its

component parts and the way in which they are put together—or a non-compositional combination—the meaning of the whole may be unrelated to the meaning of the parts. In English, the sequence ‘green leaf’ is compositional, deriving its meaning from the meaning of its components parts (‘green’ and ‘leaf’). In contrast, in human language, phonological combination is non-compositional: the English word ‘candid’, although it can be separated into two other English words (‘can’ and ‘did’), has no semantic relation to the meaning of these parts. Semantic compositionality is addressed in more depth in Chapter XX of this volume.

Several further points are merited on the meaning of complex signals. First, when a complex signal is compositional, meaning is typically derived from the *denotations* of the component parts; any pragmatic inferences are derived at the level of the complex signal as a whole when composition is complete. Second, we observe that complex signals, whether compositional or non-compositional, may provide apparent counterexamples to our definition of semantic denotation, which required meaningful features to be constant across all contexts of use. For example, the compositional sentence ‘There’s no leopard’ contains the word ‘leopard’ but is uttered when there *isn’t* a leopard in the context (due to the meaning of ‘no’); likewise, the non-compositional word ‘catapult’ contains the word ‘cat’ but will generally not be associated with the presence of a cat. The extensional definition of the semantic denotation above should thus be taken to apply at the level of a fully composed signal (for humans, roughly ‘sentences’). Once again, there is no magic formula for determining what counts as a full composed signal; one must evaluate multiple hypotheses, taking into account the role of syntax, semantics, and pragmatics (cf. Schlenker et al. 2016; Berthet et al. 2023).

6. Conclusion

The study of animal semantics presents several challenges. While the meaning of an animal signal may involve external events, signaller traits, transient states, and receiver behaviours, it is difficult to disentangle semantic denotation (the stable meaning) from pragmatic inferences (the context-dependant meaning). This can be addressed using game-theoretic principles, including notions of informativity and urgency. Several commonly-used ethological methods focus on receiver behavior; however, while this is generally a consistent inference of a signal, it is not necessarily its denotation. Overall, we encourage researchers to study the inferences of a message (including receiver behavior) *in context*, using a theory for the interaction of meaning and context, to determine the denotation.

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